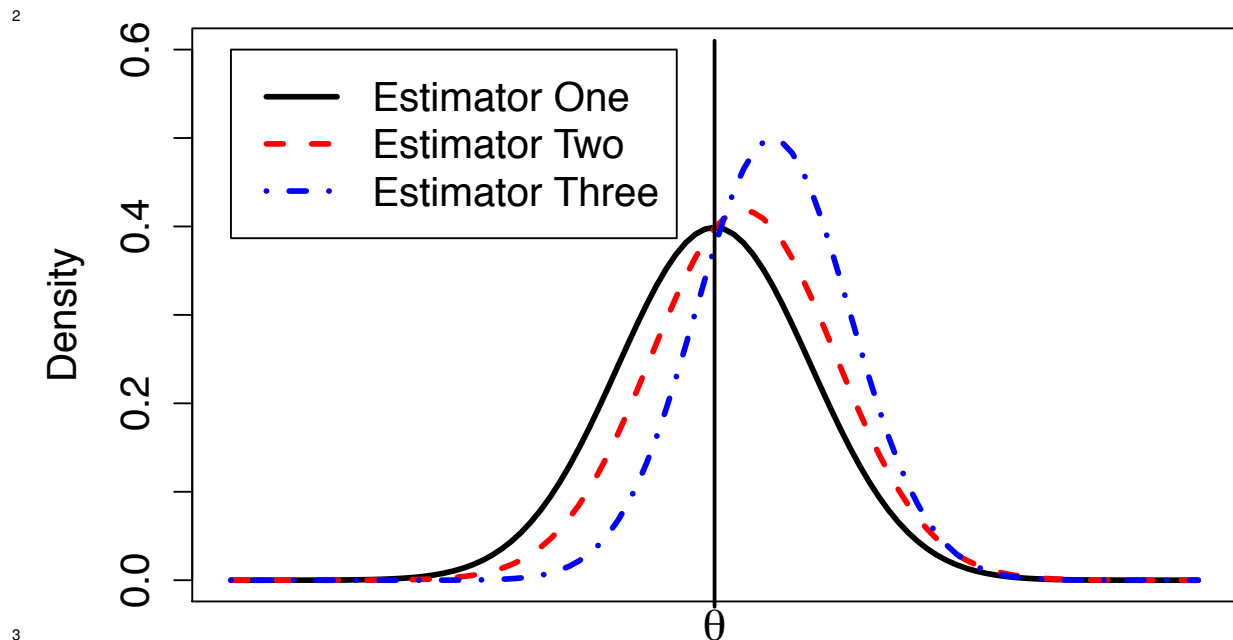


- 1 In the example below, all three estimators have the same MSE.



## 1 Important Cases

- 2 There are a few estimation situations that worth special attention.

- 3 **Case 1:** The first we have already explored, that of estimating a population  
 4 proportion. We found that the sample proportion is unbiased and has a  
 5 standard error of

6 
$$\sqrt{\frac{p(1-p)}{n}}$$

- 7 Of course, we do not know the value of  $p$ , so typically it is replaced by its  
 8 best estimate and the approximate SE is reported as

9 
$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- 1 **Exercise:** We have derived the mean and variance of the sample propor-  
 2 tion, but what is its distribution?

3  $\hat{p}$  is approx. Normal by CLT  
 4  $\hat{p} = \frac{\sum X_i}{n}$  where  $X_i = \begin{cases} 1, & \text{when } i^{\text{th}} \text{ trial is "success"} \\ 0, & \text{" " " " "failure"} \end{cases}$   
 5  $X_1, X_2, \dots, X_n$  iid Bernoulli( $p$ )

7 This approx. improves as  $n$  grows. Also better  
 8 when  $p \approx 0.5$

10 Rule of thumb: Approx. acceptable if  $np \geq 5$   
 an  $n(1-p) \geq 5$

- 1 **Case 2:** Suppose that  $X_1, X_2, \dots, X_n$  are random variables and  $E(X_i) = \mu$   
 2 and  $V(X_i) = \sigma^2$  for all  $i$ . Further assume that these random variables  
 3 are uncorrelated (meaning that every pair of random variables is uncorre-  
 4 lated).

- 5 Then, the standard estimator for  $\mu$  is the sample mean,  $\bar{X}$ , and the standard  
 6 estimator for  $\sigma^2$  is the sample variance,

$$s^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{(n-1)}$$

- 8 Note that the case where  $X_1, X_2, \dots, X_n$  are i.i.d. is a special instance of  
 9 this situation.

- 1 **Exercise:** Derive the mean, variance, and standard error of the sample  
2 mean. What is its MSE as an estimator of  $\mu$ ?

$$E(\bar{X}) = E\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n} \sum E(X_i) = \mu$$

3  $\bar{X}$  is an unbiased estimator for  $\mu$

$$V(\bar{X}) = V\left(\frac{1}{n} \sum X_i\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

6 SE of  $\bar{X}$  is  $\sigma/\sqrt{n}$ . Approx. using  $s/\sqrt{n}$

$$\text{MSE of } \bar{X} = V(\bar{X}) + \text{bias}^2(\bar{X}) = \frac{\sigma^2}{n} + 0$$

- 1 **Exercise:** If we did assume that  $X_1, X_2, \dots, X_n$  are i.i.d., what could we say  
2 about the distribution of the sample mean?

3 As long as  $n$  is "large", then the CLT  
4 says  $\bar{X}$  is approx.  $N(\mu, \sigma^2/n)$

- 1 **Exercise:** As defined, the sample variance is an unbiased estimator of  $\sigma^2$ .  
 2 (The proof is not difficult, but a little tedious.)

3 Consider the following question: What choice of  $a$  minimizes the quantity

4 
$$\sum_{i=1}^n (X_i - a)^2 ?$$

5 How does this result suggest it would be a good idea to estimate  $\sigma^2$  by  
 6 dividing the sum of squares by a constant less than  $n$ ?

7 
$$\sum (X_i - a)^2 \text{ minimized when } a = \bar{X}$$

8 
$$\boxed{\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2}$$
 is an unbiased estimator of  $\sigma^2$   
 9 since  $E((X_i - \mu)^2) = \sigma^2$   
 10  
 11  
 12

1 Note that  $\boxed{\frac{1}{n} \sum (X_i - \bar{X})^2} \leq \frac{1}{n} \sum (X_i - \mu)^2$   
 2

3 If I use  $\frac{1}{n} \sum (X_i - \bar{X})^2$  as an estimator  
 4 for  $\sigma^2$ , I will consistently underestimate  
 5 the pop. variance  
 6

7 would be better to use

8 
$$\frac{1}{k} \sum (X_i - \bar{X})^2$$
 where  $k < n$ . Turns  
 9  
 10

11 out that  $k = n - 1$  is unbiased.  
 12

- 1 **Case 3:** Suppose that  $X_1, X_2, \dots, X_n$  are i.i.d. with the normal distribution  
2 with mean  $\mu$  and variance  $\sigma^2$ .
- 3 Of course, this is a special instance of “Case 2” presented above, and the  
4 sample mean and sample variance remain unbiased estimators of  $\mu$  and  
5  $\sigma^2$ , respectively.
- 6 We can make some stronger statements, as well.
- 7 First, we know that  $\bar{X}$  has **exactly** the normal distribution with mean  $\mu$   
8 and variance  $\sigma^2/n$  for **any** sample size.
- 9 Further results depend on properties of probability distributions that we  
10 will present now.

- 1 The **Chi-squared distribution with  $k$  degrees of freedom** is a special case  
2 of the Gamma distribution. In particular, if  $X$  has the Gamma( $k/2, 1/2$ )  
3 distribution, then  $X$  has the Chi-square distribution with  $k$  degrees of free-  
4 dom.  $k = 1, 2, \dots$

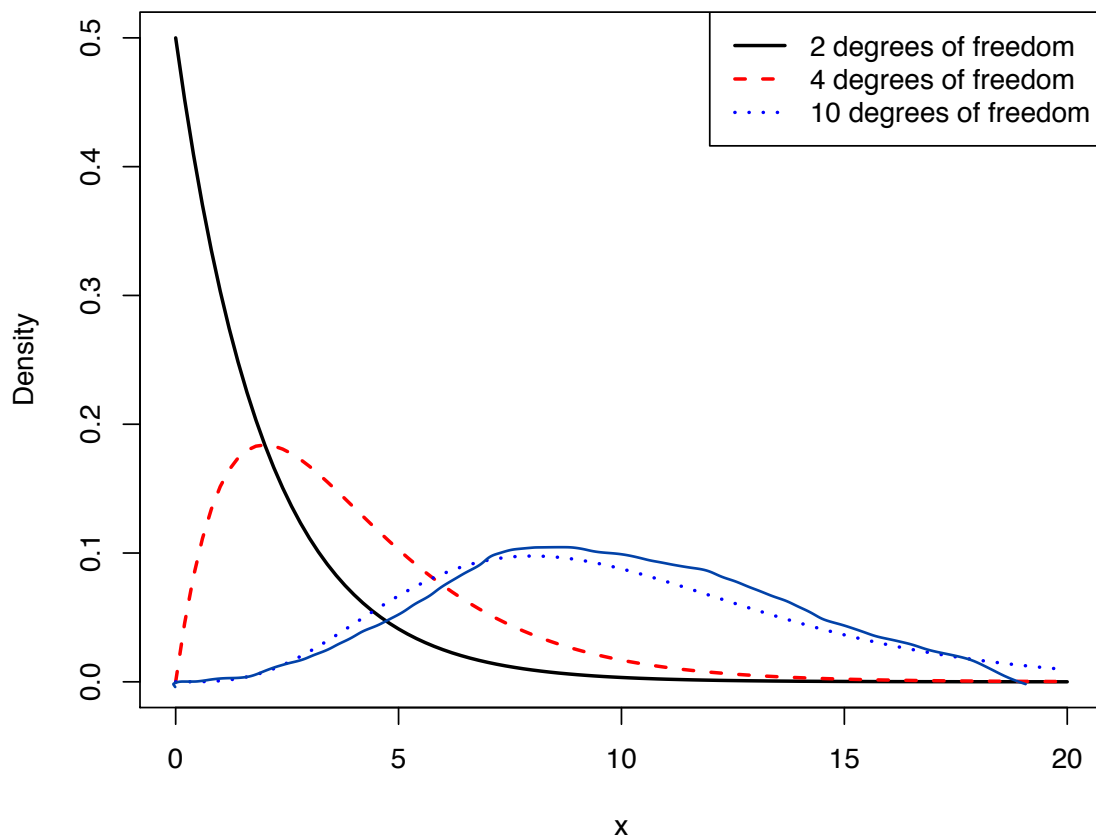
- 5 Also, if  $Z_1, Z_2, \dots, Z_k$  are i.i.d. standard normal random variables, then

6 
$$X = \sum_{i=1}^k Z_i^2$$

- 7 has the Chi-squared distribution with  $k$  degrees of freedom.

- 8 If  $X$  has this distribution, then  $E(X) = k$  and  $V(X) = 2k$ .

- 9 Plots of pdfs from the Chi-squared distribution are shown on the following  
10 slide.



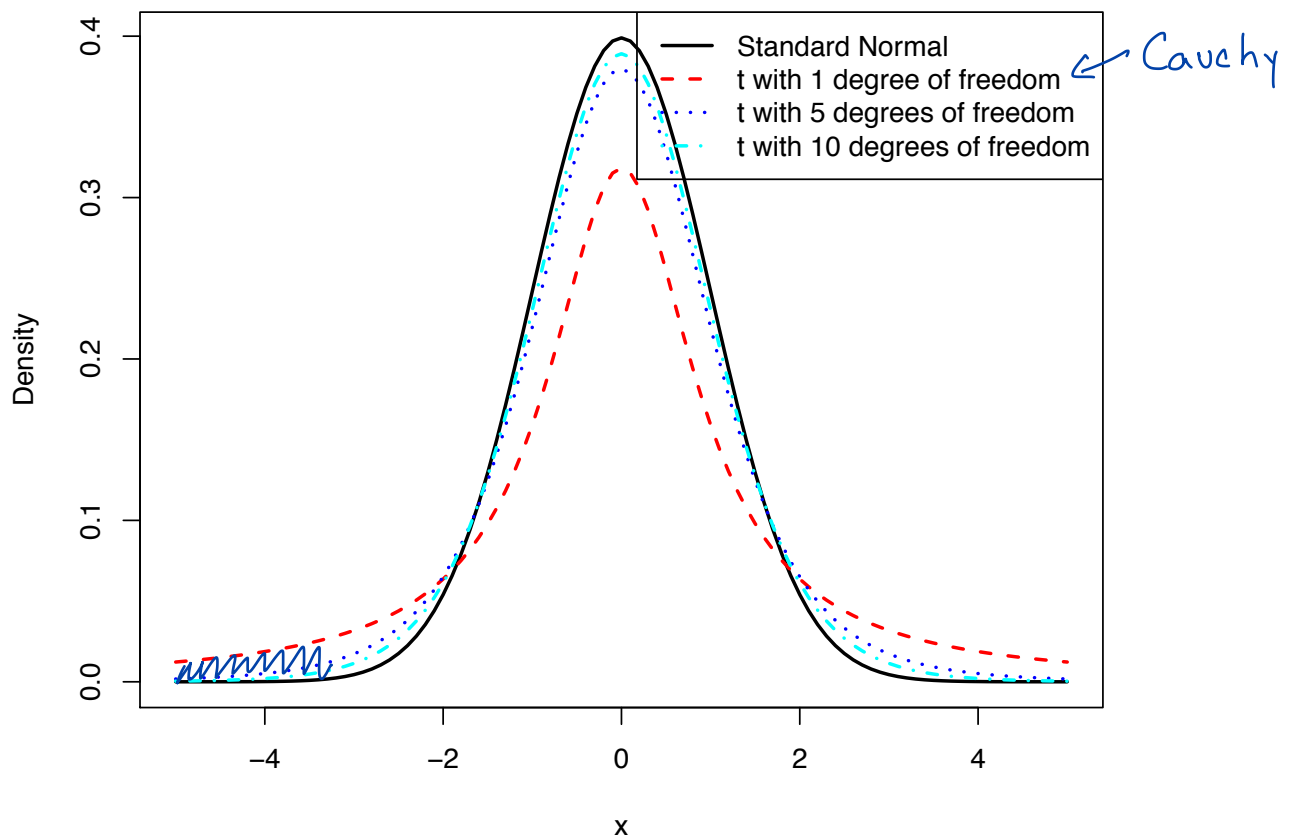
1 The  $t$ -distribution with  $\nu$  degrees of freedom is defined to be a random  
 2 variable  $T$  that can be constructed as

3 
$$T = \frac{Z}{\sqrt{X/\nu}}$$

4 where  $Z$  is a random variable with the standard normal distribution, and  
 5  $X$  is a random variable with the Chi-squared distribution with  $\nu$  degrees  
 6 of freedom. Further,  $Z$  and  $X$  are assumed independent.

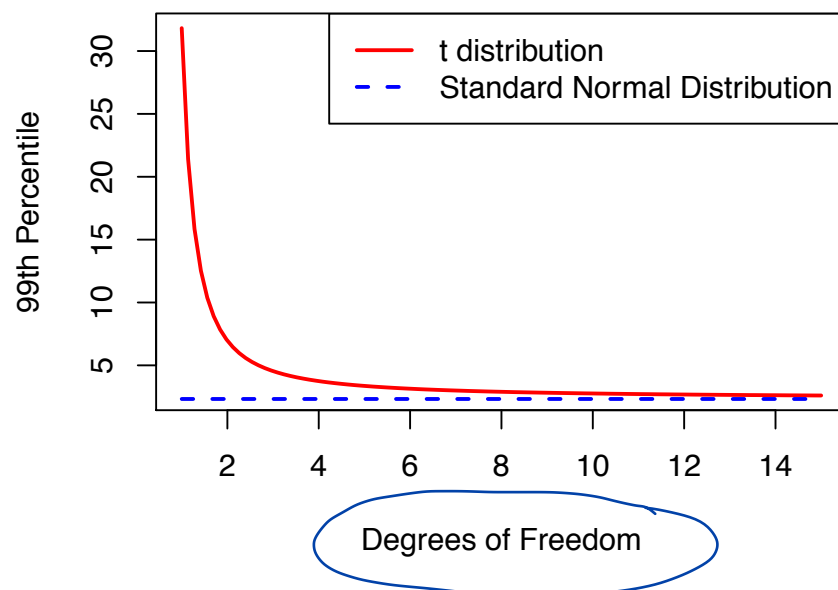
7 This distribution has a “normal-shaped” pdf, but has heavier tails than  
 8 the normal distribution. As  $\nu$  increases, the distribution converges to the  
 9 standard normal distribution.

10 It holds that  $E(T) = 0$  when  $\nu \geq 2$  and that  $V(T) = \nu/(\nu - 2)$  when  $\nu \geq 3$ .



1

- 1 The figure below makes clearer how much additional probability is in the  
 2 tails of the  $t$ -distribution compared to the standard normal. For  $\nu = 1$ ,  
 3 1% of the observations will be larger than 30. The 99th percentile of the  
 4 standard normal distribution is 2.33.



5

1 Now we can state the four **classic results** that hold under our “Case 3:”

2 Assuming that  $X_1, X_2, \dots, X_n$  are i.i.d.  $\text{Normal}(\mu, \sigma^2)$ , then

3 1.  $\bar{X}$  and  $s^2$  are independent.

4 2. The quantity

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

6 has the standard normal distribution.

7 3. The quantity

$$\frac{(n-1)s^2}{\sigma^2}$$

9 has the Chi-squared distribution with  $n-1$  degrees of freedom.

10 4. The quantity

$$\frac{\bar{X} - \mu}{s / \sqrt{n}}$$

12 has the  $t$ -distribution with  $n-1$  degrees of freedom.

1 **Exercise:** Under “Case 3” it is easier to show that  $s^2$  is an unbiased estimator  
 2 of  $\sigma^2$ , and its also easy to derive the variance of this estimator. Do these  
 3 now.

$$\begin{aligned} E(s^2) &= E\left(\frac{\sigma^2}{(n-1)} \frac{(n-1)}{\sigma^2} s^2\right) = \frac{\sigma^2}{n-1} E\left(\frac{n-1}{\sigma^2} s^2\right) \\ &= \frac{\sigma^2}{n-1} (n-1) = \sigma^2 \end{aligned}$$

↖  $\chi^2_{n-1}$

$$\begin{aligned} V(s^2) &= V\left(\frac{\sigma^2}{(n-1)} \frac{(n-1)}{\sigma^2} s^2\right) = \frac{\sigma^4}{(n-1)^2} V\left(\frac{n-1}{\sigma^2} s^2\right) \\ &= \frac{2\sigma^4}{(n-1)} \end{aligned}$$

11 The SE of  $s^2$  is  $\sqrt{\frac{2\sigma^4}{n-1}}$



## 1 Confidence Intervals

2 The already-mentioned standard error (SE) is a useful tool for quantifying  
 3 error in an <sup>estimator</sup> estimate, but a common alternative is the use of **confidence**  
 4 **intervals** for unknown parameters.

5 Formally, we call  $(L, U)$  a **100(1 -  $\alpha$ )% confidence interval for  $\theta$**  if

6

$$P(L \leq \theta \leq U) = 1 - \alpha$$

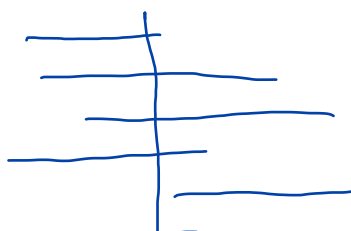
7 Hence, if want a “95% confidence interval,” set  $\alpha = 0.05$ . Other common  
 8 choices are 90% and 99% confidence intervals ( $\alpha = 0.10$  and  $\alpha = 0.01$ , re-  
 9 spectively).

1 An appropriate statement regarding a confidence interval is, for example,  
 2 “I am 95% confident that the true value of  $\theta$  is between 10.6 and 11.7.”

3 It is **not** appropriate to say that “The probability that  $\theta$  is between 10.6  
 4 and 11.7 is 0.95.” This is incorrect because there is nothing random in that  
 5 statement:  $\theta$ , 10.6 and 11.7 are all constants.

6 The randomness is in the procedure itself. If one constructs 95% confidence  
 7 intervals in a wide range of situations, then 95% of those intervals will  
 8 cover the true value of the parameter.

9 There is an alternative approach to statistical inference, **Bayesian inference**,  
 10 in which parameters are treated as random variables. More on that later.



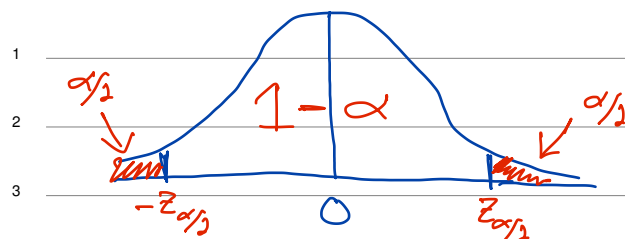
- 1 **Exercise:** Use our “classic results” above for “Case 3” to derive confidence  
 2 intervals for  $\mu$  and  $\sigma^2$  in that situation. For estimating  $\mu$ , consider both the  
 3 instances where  $\sigma^2$  is known and unknown.

4 ① CI for  $\mu$ ,  $\sigma^2$  is known

5 know that  $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$

6 Thus,  $P\left(\underline{-z_{\alpha/2}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \underline{z_{\alpha/2}}\right) = 1 - \alpha$

7 where  $z_a$  is the point under the standard  
 8 normal dist. such that the area above  
 9 equals  $a$



5  $P\left(\underbrace{\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}_L \leq \underbrace{\mu}_{\text{circled}} \leq \underbrace{\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}_U\right) = 1 - \alpha$

6  
7  
8  
9 So,  $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$  is a  $100(1-\alpha)\%$

10 CI for  $\mu$

② Estimating  $\mu$ ,  $\sigma^2$  unknown

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$P(-t_{\alpha/2, n-1} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{\alpha/2, n-1}) = 1 - \alpha$$

Find  $100(1-\alpha)\%$  CI for  $\mu$  is

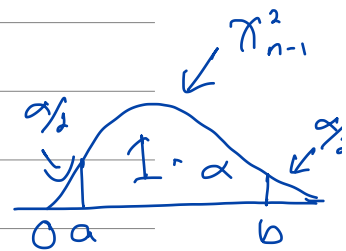
$$\bar{X} \pm t_{\alpha/2, n-1} s/\sqrt{n}$$

Note that  $\underbrace{t_{\alpha/2, n-1}} > z_{\alpha/2}$

③ CI for  $\sigma^2$

$$\text{Use } \frac{(n-1)}{\sigma^2} s^2 \sim \chi^2_{n-1}$$

$$P(a \leq \frac{(n-1)s^2}{\sigma^2} \leq b) = 1 - \alpha$$



$$P\left(\underbrace{\frac{(n-1)s^2}{b}}_L \leq \sigma^2 \leq \underbrace{\frac{(n-1)s^2}{a}}_U\right) = 1 - \alpha$$

So,  $\left(\frac{(n-1)s^2}{b}, \frac{(n-1)s^2}{a}\right)$  is the desired CI

1 The next slides present four important confidence intervals that are com-  
2 monly used.

3 *Confidence Interval for Population Mean,  $\mu$ , when  $n$  is large ( $n \geq 30$ )*

4 Assume that the sample is i.i.d. from a distribution with mean  $\mu$  and vari-  
5 ance  $\sigma^2$ , with both  $\mu$  and  $\sigma^2$  unknown. Then, a  $100(1 - \alpha)\%$  confidence  
6 interval for  $\mu$  is

$$\bar{X} \pm z_{\alpha/2} \left( \frac{s}{\sqrt{n}} \right)$$

7  
8 where  $z_a$  is such that  $P(Z > z_a) = a$  when  $Z$  has the standard normal  
9 distribution. ~~In R notation,  $z_a$  equals  $qnorm(1-a)$ .~~

1 *Confidence Interval for Population Mean,  $\mu$ , when  $n$  is small ( $n < 30$ )*

2 Assume that the sample is i.i.d. from the normal distribution with mean  
3  $\mu$  and variance  $\sigma^2$ , with both  $\mu$  and  $\sigma^2$  unknown. Then, a  $100(1 - \alpha)\%$   
4 confidence interval for  $\mu$  is

$$\bar{X} \pm \underline{t_{\alpha/2, n-1}} \left( \frac{s}{\sqrt{n}} \right)$$

5  
6 where  $t_{a,v}$  is such that  $P(T > t_{a,v}) = a$  when  $T$  has the  $t$ -distribution with  
7  $v$  degrees of freedom. ~~In R notation,  $t_{a,v}$  equals  $qt(1-a, v)$ .~~

# 1 Confidence Interval for Population Variance, $\sigma^2$

2 Assume that the sample is i.i.d. from the normal distribution with mean  
 3  $\mu$  and variance  $\sigma^2$ , with both  $\mu$  and  $\sigma^2$  unknown. Then, a  $100(1 - \alpha)\%$   
 4 confidence interval for  $\sigma^2$  is

$$5 \left( \frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{(1-\alpha/2), n-1}^2} \right)$$

6 where  $\chi_{a,v}^2$  is such that  $P(U > \chi_{a,v}^2) = a$  when  $U$  has the chi-squared  
 7 distribution with  $v$  degrees of freedom. ~~In R notation,  $\chi_{a,v}^2$  equals~~  
 8  ~~$\text{qchisq}(1-a, v)$ .~~

# 1 Confidence Interval for Population Proportion, $p$

2 Assume that  $n$  is large enough that you are confident that  $np \geq 5$  and  
 3  $n(1-p) \geq 5$ . Let  $\hat{p} = X/n$  where  $X$  is binomial( $n, p$ ), and  $p$  is unknown.  
 4 Then, a  $100(1 - \alpha)\%$  confidence interval for  $p$  is

$$5 \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

6 where  $z_a$  is such that  $P(Z > z_a) = a$  when  $Z$  has the standard normal  
 7 distribution. ~~In R notation,  $z_a$  equals  $\text{qnorm}(1-a)$ .~~

$$\frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \sim N(0,1)$$

sp.norm.ppf(.975, 0, 1)  
 returns 1.96

1 **Exercise:** A sample of size 50 is drawn from a population with unknown  
2 mean and variance. It holds that  $\bar{x} = 14.6$  and  $s = 1.3$ . Estimate the  
3 population mean. State both the standard error for the estimator and a  
4 90% confidence interval for  $\mu$ .

5 \_\_\_\_\_

6 \_\_\_\_\_

7 \_\_\_\_\_

8 \_\_\_\_\_

9 \_\_\_\_\_

10 \_\_\_\_\_

11 \_\_\_\_\_

12